

Essentials of Innovation & Entrepreneurship (EIE) - II

Module (M1): Orientation, Introduction to EIE, Why Startups Fail

Icebreaker!!

Interactive Poll (Mentimeter)



Join at
www.menti.com

Use the code: 1436 7437



Meet the Team



Prof. Sathya Prasad
Lead Instructor, CIE@RR
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Lead Instructor, CIE@EC
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CIE Program Manager
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Mr. Raghavendra Deshmukh
Co-Instructor, RR Campus, Industry Mentor
raghavendradeshmukh@pes.edu



Mr. Tarun Rama
CIE Ignite Manager
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Department Coordinators
(Respective Departments)

E-Cell
Student body
(EC & RR campus)

Meet the Team



Chirag C Shekar

Teaching Assistant, Intern
chiragcshekar54@gmail.com



Pulkit Aggarwal

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CIE Social Media Community



[@cie.pesu](https://www.instagram.com/cie.pesu)



[Center for Innovation and Entrepreneurship @ PES University](#)



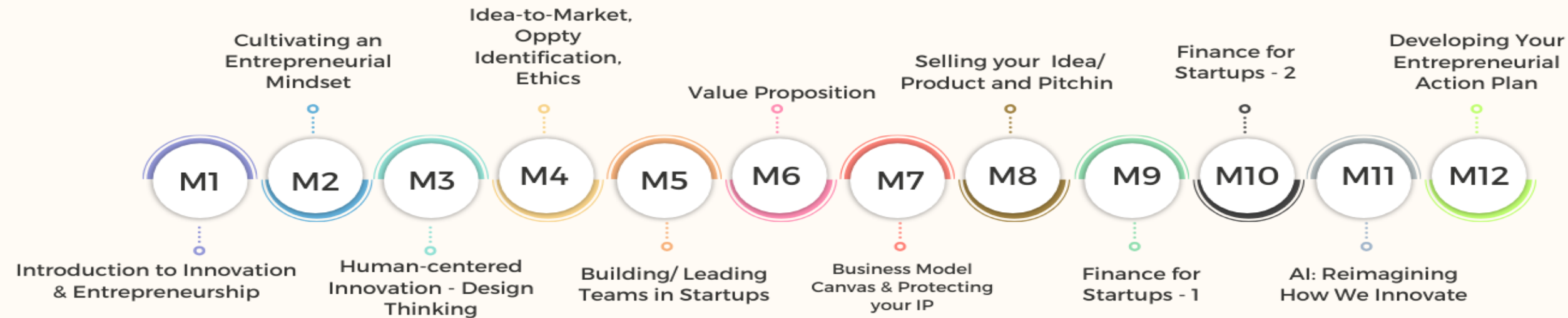
[@CIECommunity](#)



ESSENTIALS OF INNOVATION & ENTREPRENEURSHIP - I (EIE I) (AUG-DEC 2025)



Course Modules:



Course Template:

Lectures			ISA	ESA
Case Studies	Video Discussions			
In-Session Quiz	End of session Quiz	External Speakers		
Integrated Hackathon and Mentorship				

CIE Spark: Idea/Problem Validation Sprint



Guest Speakers:

Industry(9)

K. Chandrashekar
Founder, CEO – Forus Health
Srivibhavan Balaram (Vibhav)
MD, VP – Stryker, Angel Investor
Raghavendra Bhat
Principal Engineer, AI – Intel
Raghavendra Bhat
Principal Engineer, AI – Intel
Varun Udbalker
AI Engineer – 4GoodAI

Harsha Chaturvedi
CEO – 4GoodAI, CTO – PESU
Medha Koppam
Microsoft Research, AI
Rahul Srinivas
CoS – AstraFin Neosolutions
Rakesh Godhwani
Founder – School of Meaningful Experiences (SoME)

Alumni Startup(5)

Ananya Mungara
CEO – Smart Chakra
Aditya Kolavi
Founder – Cognitive Labs
Abhilash Madabushi
Founder – Consuma.ai
Shashank Prabhakar
Co-Founder – draft42, Kaadoo Trails
Aditya
Co-Founder – Team Aalaap

Faculty(2)

Dr. Chitra
Head of Design, PESU
Prasanna Chandran
Industry Mentor – PESU CIE,
Consultant – Greywiz

16 Guest Speakers
21 Disciplines

4000+
Students (RR & EC campus)

Partners



Transforming Theory Into Practice



Founder / Expert Talks

- | | | |
|--|--|--|
| PESU Alums (11) <ul style="list-style-type: none"> Tushar Magar - (Founder, Daimyo) Manoj Sanker (Founder, Nemocare) Aditya Mehta (Founder, Epicure Robotics) Abhilash Madabhushi (Founder, Consuma.AI) Shashank Prabhakar (Founder, Kaadu Trails, draft42) Dhruv Shetty (Founder, Engineer) Keerthi Raghavendra (grad student, CMU) Shriya Desai (Deloitte) Samhitha Harish (EY) Parthan M (grad student, Aachen Univ) Shreyas Vedapatak - (8thSem Mech) | Faculty Experts(7) <ul style="list-style-type: none"> Prof. Nagarjuna S (Pro-VC) - Importance of Intrapreneurship Prof. Nidhi Srivatsa (Law) - IPR for Startups Prof. Manikandan (ECE) - Idea->Patent: What & How? Prof Ravi Urs (FOMC) - Digital Marketing, Digital Biz Models Prof. Sushma (Performing Arts)- Storytelling for Startups Dr. Jaykumar (BHEM) - Why/What/How | External Experts(6) <ul style="list-style-type: none"> John Kuruvilla (CEO) - Serial Entrepreneur; multi-area Krishnaprasad Jagadish (Founder CEO, Parjanya) - AR/VR & Media: startup journey Suresh Narasimha (CEO, Co-Create Ventures, PVL) - How to succeed in student-led startups Mr. S J Vijay (CEO) - India Pioneer, Building-as-a-Product Mr. Nithin Simakurti (CEO) - Service/Product company from scratch Sushma Krishnamurthy - InzpireU |
|--|--|--|

- | | |
|---|---|
| Students <ul style="list-style-type: none"> 4000+ 21 Disciplines 2 Campus | Dept <ul style="list-style-type: none"> FOMC Depts of Engg CIE TA's Industry Mentors |
|---|---|

CIE Ignite Jury

- Dr Sudeendra Koushik** BE, MBA, PG SM (IIM-K), PhD (INNOVATION).
- Bindu S. Sastry** Founder, MD, MAUVERICK
- Anand Sri Ganesh** CEO - NSRCEL, IIM Bangalore

CIE Ignite Sponsors

- Title Sponsor - **CiC (Co-Innovation Center by PESU & IHFC@IITD)**
- Platinum Sponsor - **InzpireU**
- Gold Sponsor - **4GoodAi**
- Category Sponsor - **SL Memorial**

CIE Ignite (Ideathon)



Stages	Preliminary Pitch(S1)	Team Pitch(S2)	Grand Finale(S3)
#Teams	484	94	10
Participants	3000	564	55

Grand Finale - 11th April 2025

Jan-April 2025

Total Prize Money - ₹70,000

- Winner - ₹25,000
- 1st Runners up - ₹15,000
- 2nd Runners Up - ₹10,000

Category Awards

- Prototype Pioneer Award - ₹5,000
- Blockbuster Visionary Award - ₹5,000
- All Girls Team - ₹5,000
- People Choice Award - ₹5,000

Course Template				
EIE II - 4 th Sem			ISA	ESA
Lectures				
Case Studies	Video Discussions			
In-Session Quiz	End of session Quiz	External Speakers		

Course Outline*

	Theme/Topic	CIE-Ignite Milestone*
M1	Orientation, Introduction to EIE, Why Startups Fail	
M2	Market Failure, Validation and Pretotyping	
M3	Customer Development (CD): Hypothesis & Assumptions	Stage0 (S0)
M4	Customer Development (CD): Interviews, Evidence	
M5	Customer Development (CD): Experiments & Customer loops	
M6	Startup Pitching	Stage1 (S1)
M7	Innovation Framing, Jobs-To-Be-Done (JTBD)	
M8	Marketing and Market Structure	Stage2 (S2)
M9	Startup Finance 201	
M10	Startup Finance 201 (Part 2)	Stage3 (S3)
M11	IP for Startups	
M12	Reflections, Synthesis, New Journey	

* Proposed, subject to modification

How Can EIE Help You in Your Education/Career?

We envision this to be more than a 'course':

- Discover your 'True North' / Interests / Passion
- Identify ideas/opportunities that you consider worth pursuing (tech/biz / startup/research / etc)
- Find potential collaborators/teammates/startup founders/innovators
- Get funding/prototyping/mentoring / etc support to build out your idea/team

Sample/Illustrative Example: One possible path (*not* the only path)



Build Tech, Business and Soft skills; Discover their 'True North'/passion, Teammates and more (build prototype/minimum-viable-product, validate, etc.)

Apply to CIE PESU-IITx Co-Innovation-Center (grant, mentorship) Corporate/Govt Incubator/Accelerator, Research Labs, etc.

* Faculty & Students

Course Objectives

- Prepare you to kickstart your I&E Journey *now!*
- Provide a platform to apply your I&E skills (CIE Ignite)
- Exposure to transformative technology-driven innovations
- Interactions with Founders/VCs/Startup Ecosystem experts
- Deeper awareness and understanding of business acumen

**Fun way to prepare for an exciting/impactful
3rd/4th year PESU journey!**



Learning Outcomes

- Applying what you learned in **EIE-1** (BMC, VPC, Design Thinking, etc)
- Opportunity for you to (safely) create a startup by picking a real-world problem (opportunity) & solving it using proven principles
- Use Lean Startup & Customer Development principles for your idea
- “Evidence-based” vs. “faith-based” entrepreneurship
- Discover great teammates – potential/future Co-Founders!

How will you apply EIE learnings with your other Courses/Projects/Gig(s)?



Course Mechanics



Credits: 2 Credits (Weekly 2hr)

Engagement: Mostly Online (some sessions will be F2F with advance notice)

Will include talks by Startup Founders, Industry Experts, PESU Faculty, etc

Grading*: S: 90 –100 A: 80 –89 B: 70.-79 C: 60 –69 D: 50 –59 E: 40 –49 F: 39 & below

Assessment*: [CIE Ignite/ideathon](#)(10%), CBT - 2 ISAs (20% each) & 1 ESA (50%)

Attendance: **Same as university requirements (75%+)**

Dept Support: Contact the respective Depts Coordinator (DC) for assistance (see next slide)

References:

- Technology Ventures: From Idea to Enterprise, McGraw Hill
- PES University 'CIE Level 1' Course - [Getting Started With Entrepreneurship](#)
- VentureLab ([Entrepreneurial Skillset and Mindset](#))
- European Union (EU) [Entrepreneurship Competencies](#)

* Proposed, subject to modification



EIE Course Department Coordinators

Dept	Chairperson	Faculty DC	Mail ID
AIML	Dr. Jayashree R	Prof. Shwetha	shwethakn@pes.edu
BArch	Prof. Hemashree	Prof. Yashas Hegde	yashashegde@pes.edu
BA - Performing Arts	Dr Sushma S V	Ms. Prapujya Ramesh	prapujyaramesh@pes.edu
BCA	Dr. Veena S	Prof. Santosh Katti	santosh_katti@pes.edu
BDes	Prof. Chitra Vishwanathan	Prof. Smriti Haritz	smrthiharits@pes.edu
Biotech	Dr. Reshma	Dr. Jhinuk Chatterjee	jhinukchatterjee@pes.edu
CS	Dr. Shylaja S	Dr Gayathri T	gayathrit@pes.edu
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EEE	Dr. Rex Joseph	Dr. Sangeeta Modi	smodi@pes.edu
Mech	Dr. Rajesh Mathivanan	Dr. Kiran M	kiranm@pes.edu
BA LLB	Prof. Jomy George	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu
BBA LLB	Prof. Jomy George	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu
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BBA Bus Analytics	Dr. Jayashree Sapra	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu
BBA HEM	Dr. Jaykumar V	Prof. Bhaskar Sailesh	bhaskars@pes.edu
Bcom General	Dr Anuradha Pai	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu
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BSc Economics	Dr Anuradha Pai	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu
BSc Psychology	Prof. Samini Mathew	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu
Intl Accts & Finance	Dr. Shailashree Haridas	Prof. Sangeetha Kumar	sangeethaskumar@pes.edu

Session Quiz - FAQ

For a smooth Session Quiz experience, pls follow the below guidelines:

✓ Session-End Quiz – Instructions

1 Log in to **PESU Academy** and navigate to **My Quizzes** – you will find your test details there.

2 This will take you to the **Assessment Page**. Use the following credentials:

- 🔑 **PRN:** Your PRN Number
- 🔑 **Password:** Your PESU Academy password

3 If you are unable to log in to the assessment page, please email **support@pesuacademy.com** with the following details:

- ✓ SRN
- ✓ Name
- ✓ Screenshot of the My Quizzes tab at the time of the issue

4 If no test appears in **My Quizzes**:

- Log in at www.pesuacademy.com/Assessment
- Check if the message shows "*No Test Scheduled*"
- Take a screenshot as proof

How to Get the Best Out of this Course?



- Discipline, Perseverance & Commitment. (key ingredients of a successful entrepreneur!)
- Practice the framework/tools: Leverage CIE-Ignite (develop skills to use the 'tools'; not formulaic, find out what works best for you)
- Lots of 'doing' in a Team setting (if Team succeeds, you succeed)
- Leverage mentorship from CIE Industry Mentors for Team Project
- Use the new CIE arena and Makerspace to build your prototype
- Based on your efforts/results, apply for CIE Summer Internship

Student Engagement / Facilitation



Online

- Q/A on MS Teams (Moderated by CIE TAs)
- Polls
- Session Quiz on PESU Academy: **Important!** (this is how you get attendance)
- For course related assistance, reach out to cieoffice@pes.edu (Madhukar Sir)

Offline

- Department Coordinator (DC) in your respective Department





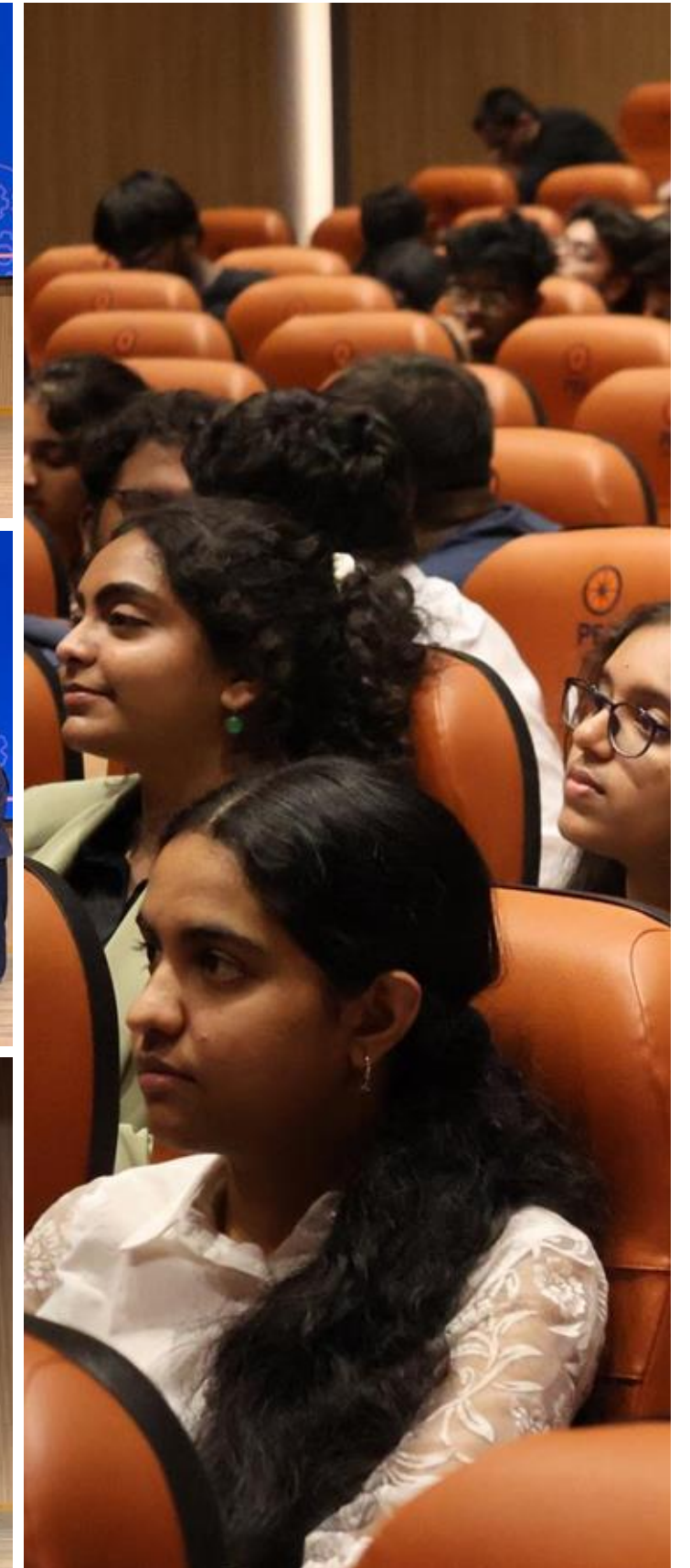
For Tomorrow's Innovators

CIE Ignite Season02: A curriculum-integrated innovation program delivering guaranteed engagement with 4,500+ engineering students at their most motivated moment.

From classroom theory to real-world impact!



Glimpse of CIE Ignite Season 01 - 2025



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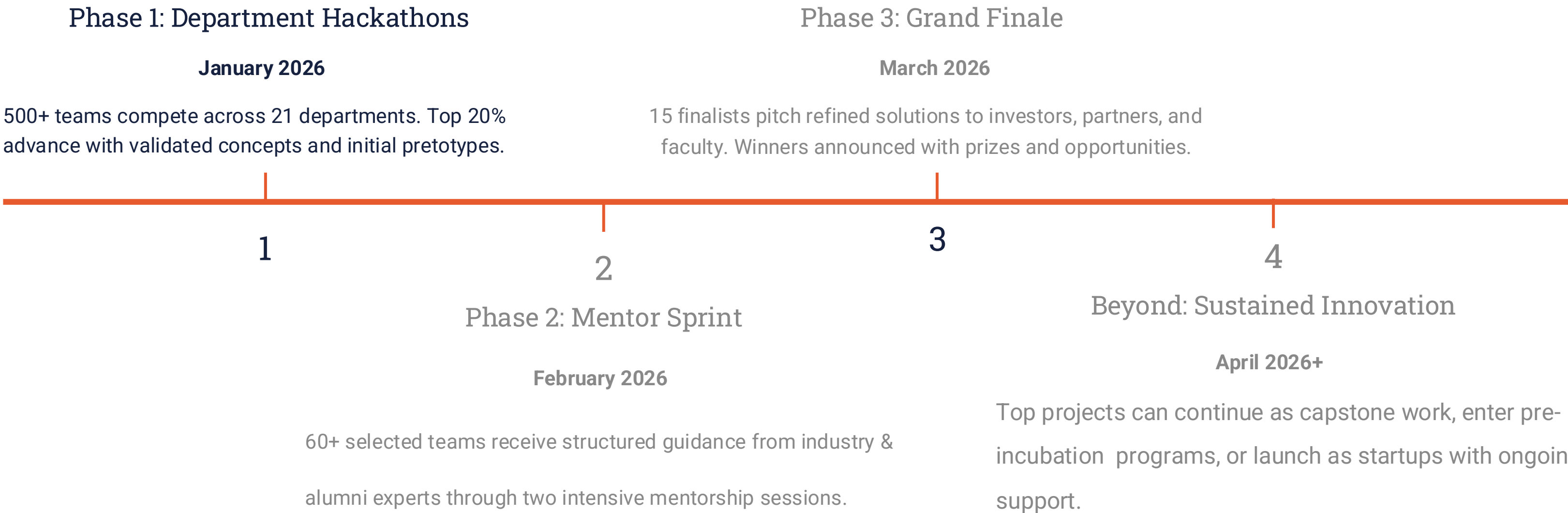
GOLD SPONSOR



KNOWLEDGE PARTNER

The CIE Ignite Journey

A three-phase innovation pipeline that transforms classroom learning into validated, market-ready solutions



Innovation starts here—and it does not stop.



Season 02 powers ideas into impact:

- Hackathons → Mentor Sprint → Grand Finale → Beyond
- Teams sharpen, validate, and pitch to real stakeholders
- Top projects continue as capstones, pre-incubation, or startups

Be part of the journey:

- Faculty: shape problem statements
- Industry: mentor, fund, and co-build
- Students: bring your best

Let us build market-ready solutions from PES!

Contact: tarunrama@pes.edu

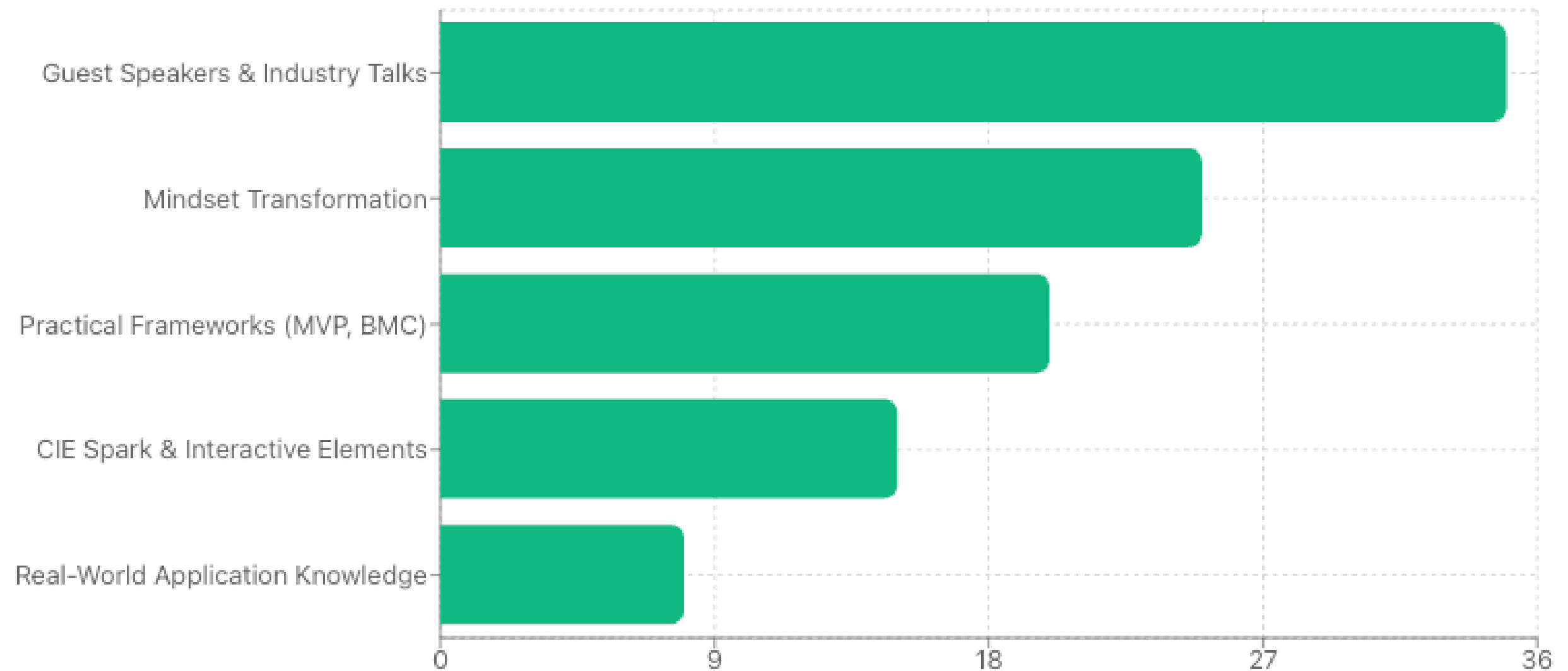
Visit: <https://cie.pes.edu/>



We Asked, You Responded

🌟 What Was a Game-Changer? (103 Students)

Top 5 Most Impactful Elements

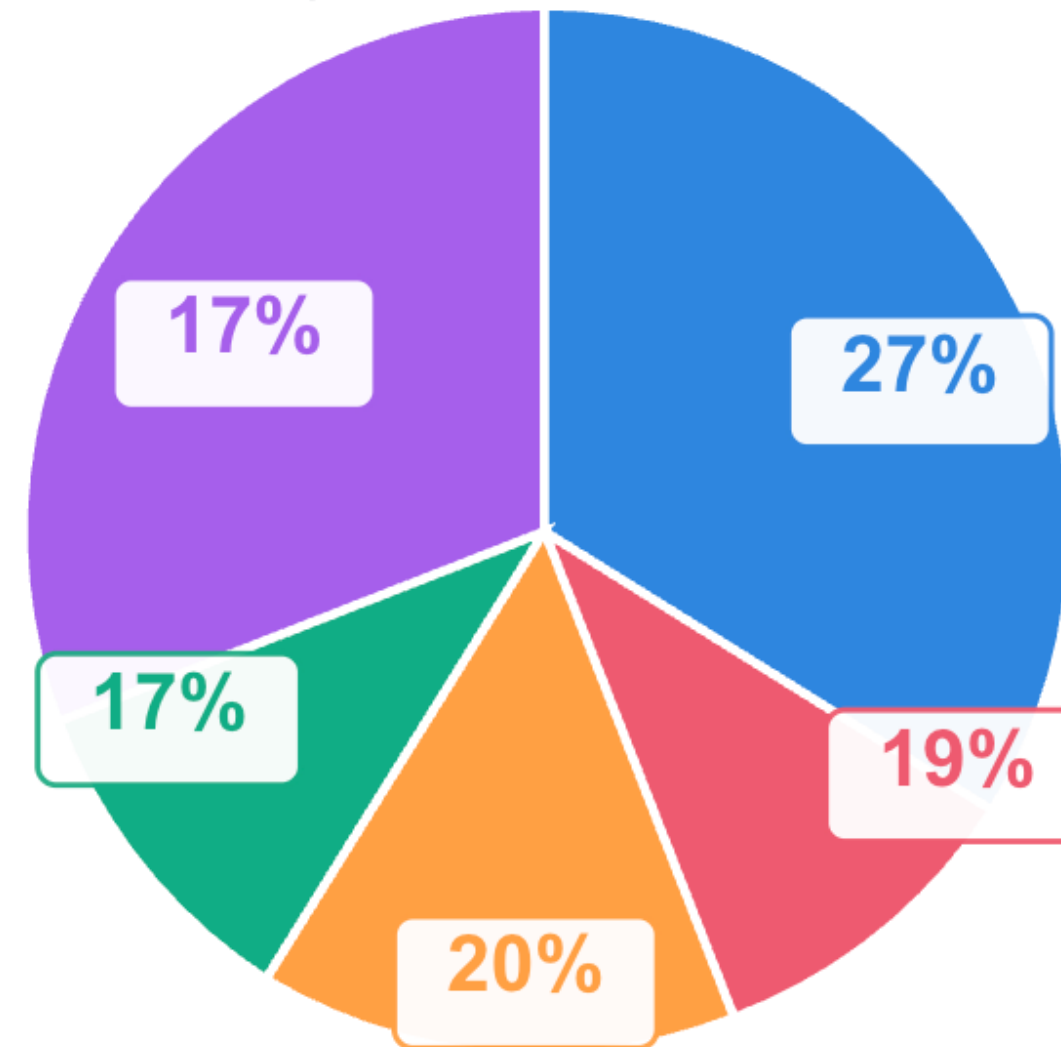


Who of you offered your PoV?



Who Gave Feedback?

Department Distribution



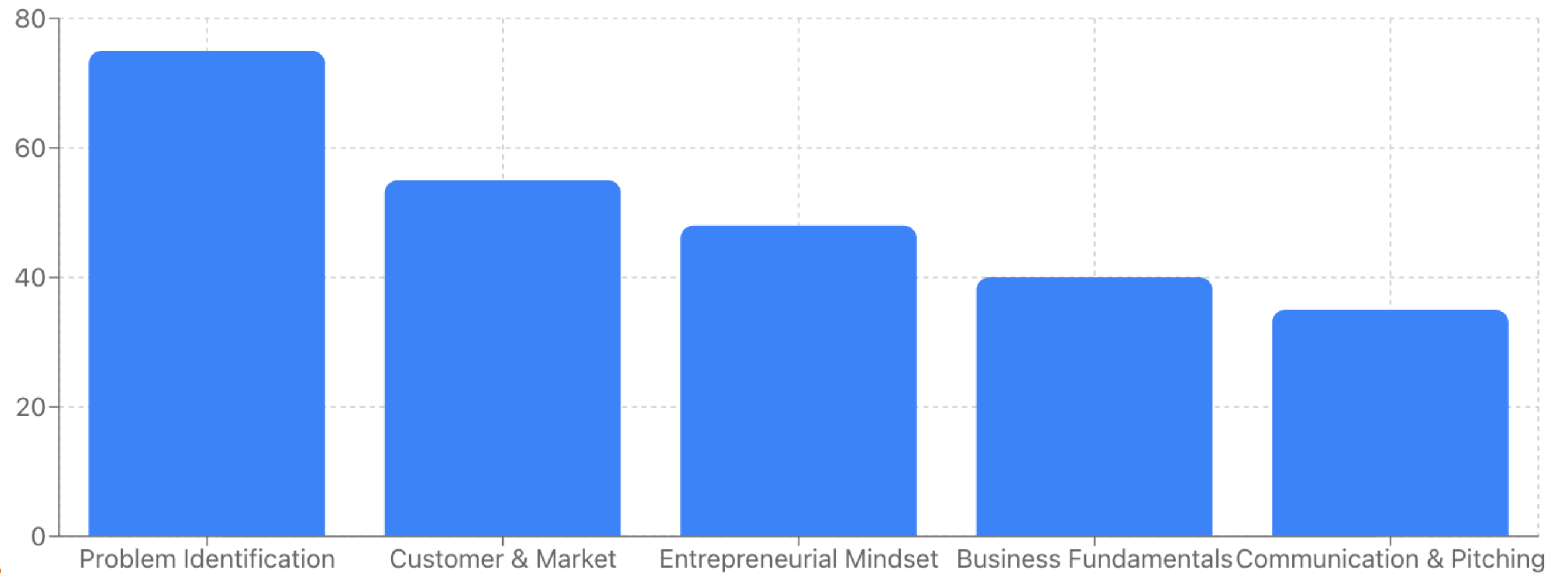
62% from Engineering (CSE/AI&ML/ECE) | 38% from Non-Tech

Your learnings from EIE1



Top Learnings from CIE

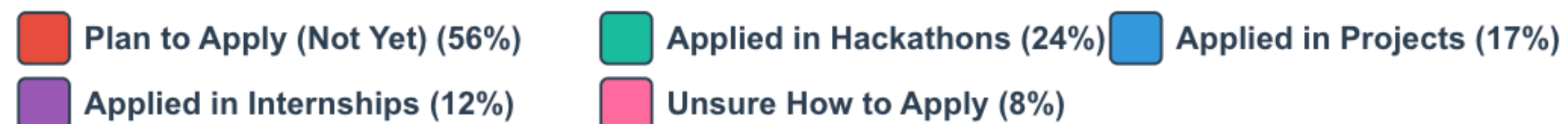
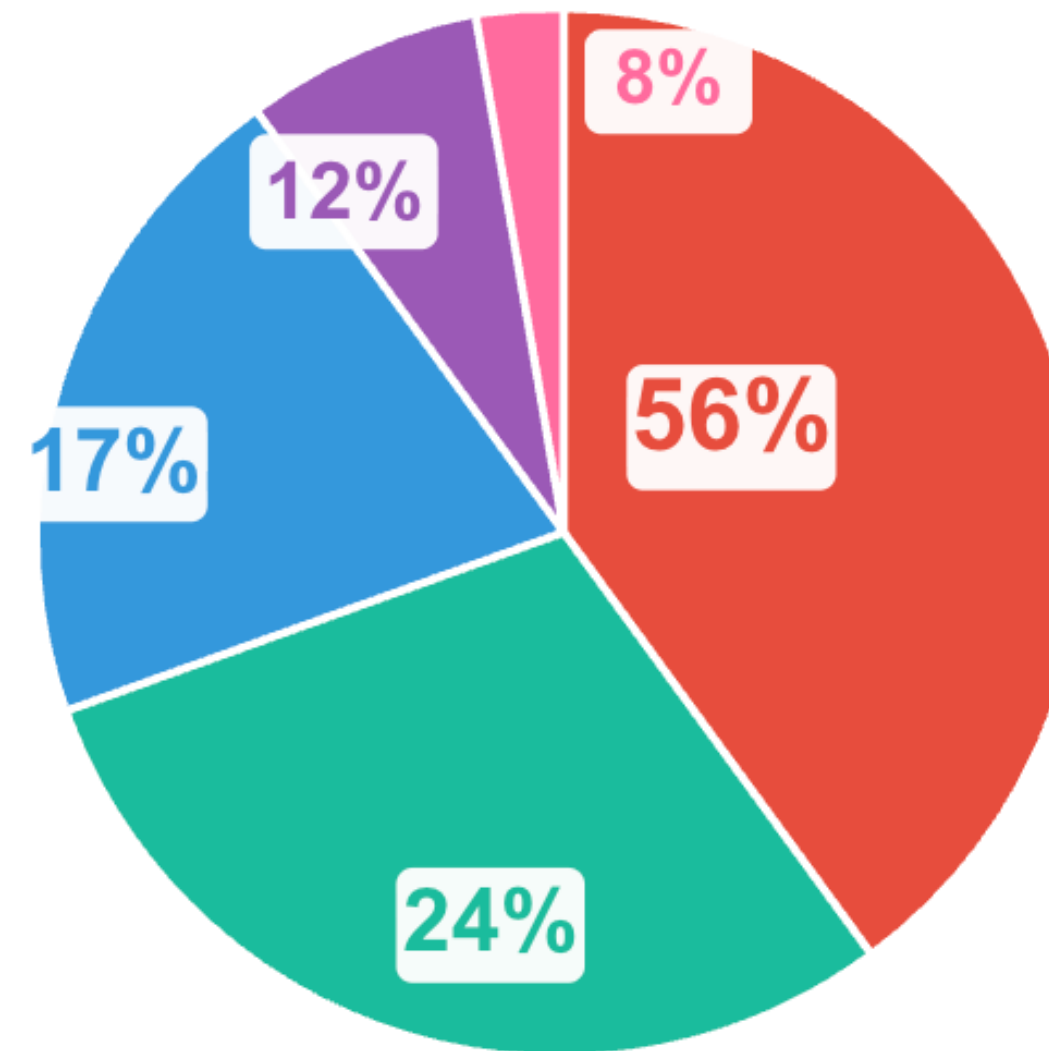
Most Frequently Mentioned Takeaways



Have you applied what you learnt?

⚠ The Application Gap

Have Students Applied What They Learned?

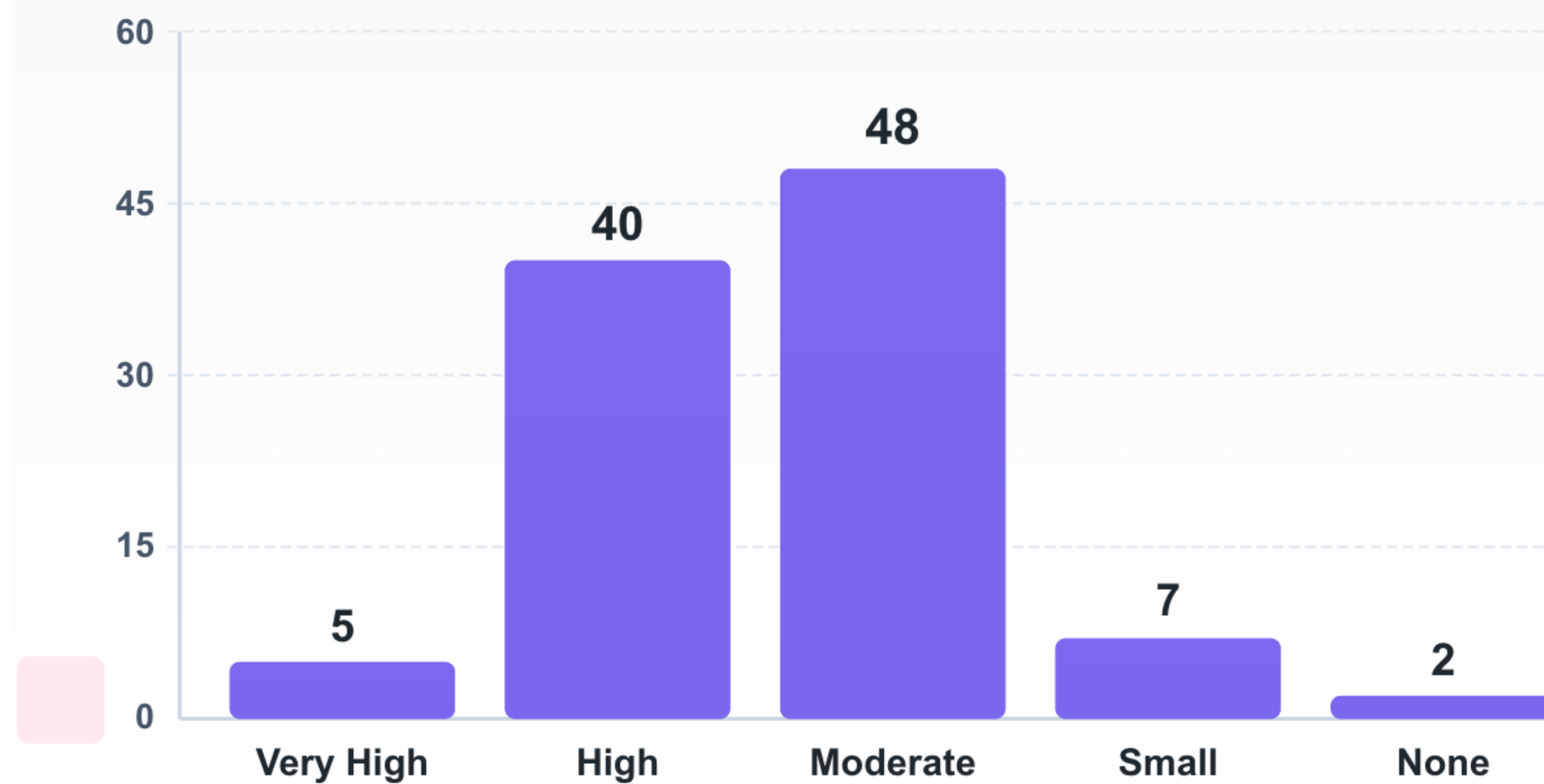


EIE Impact



Overall Impact Rating

How Much Impact Has CIE Had on You?



88 responses • 88% reported Moderate to Very High impact

EIE1 – Strengths and Gaps



Course Effectiveness Across Dimensions

Strengths and Gaps

✓ STRENGTHS

Innovation Thinking

85%



Engagement

75%



Career Shaping

70%



! GAPS TO ADDRESS

Practical Application

60%



🔴 Critical: Students struggle to implement learnings

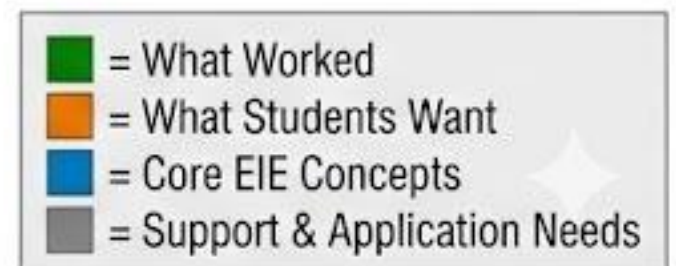
Content Depth

65%



Need deeper dive into fewer topics

Overall Signals



Theme for EIE2 – H1 2026



Engage Better

Request – Student Volunteers



<https://forms.gle/4AHHohNhhMDGoxWy6>



Let us Play – Signal OR Noise



At **CES 2026** (A Tech showcase event, Las Vegas - Jan 2026) wearable AI gadgets focused on lifelogging and personal assistance were highlighted as a standout category, with major companies (*Meta, Amazon, OpenAI* and newer startups) showing prototypes and early products.

Multiple startups unveiled wearable AI devices that record and summarize daily life (audio, visual, contextual data), helping users capture memories and automate routine tasks. Investors are pouring early capital into this space, and some product makers are already doing early consumer testing.

Your Turn: Signal vs Noise Question:

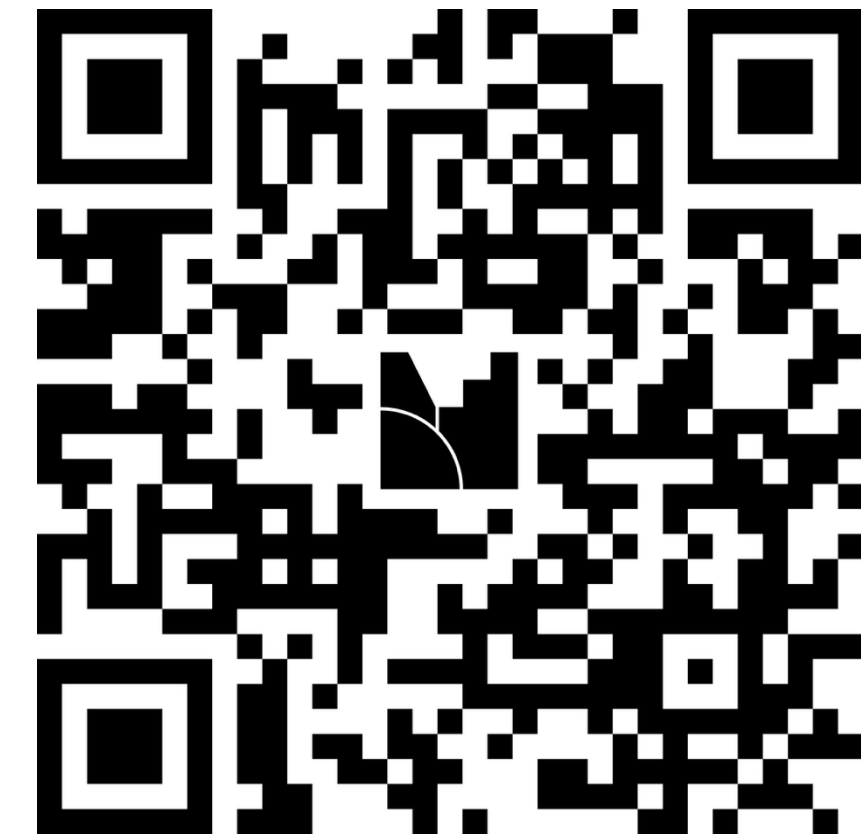
**Is wearable AI a real customer need that will scale or hype that may not stick with real users?*

You Vote using the Menti

Signal: strong market need

Noise: temporary buzz

Menti: 6844 9636



Why Startups Fail and What Can You Do Differently?



What is a Startup?

A startup is a temporary organization designed to search for a repeatable and scalable business model



Why startups fail?

Most Startups want to build this



Why startups fail?

First build this....

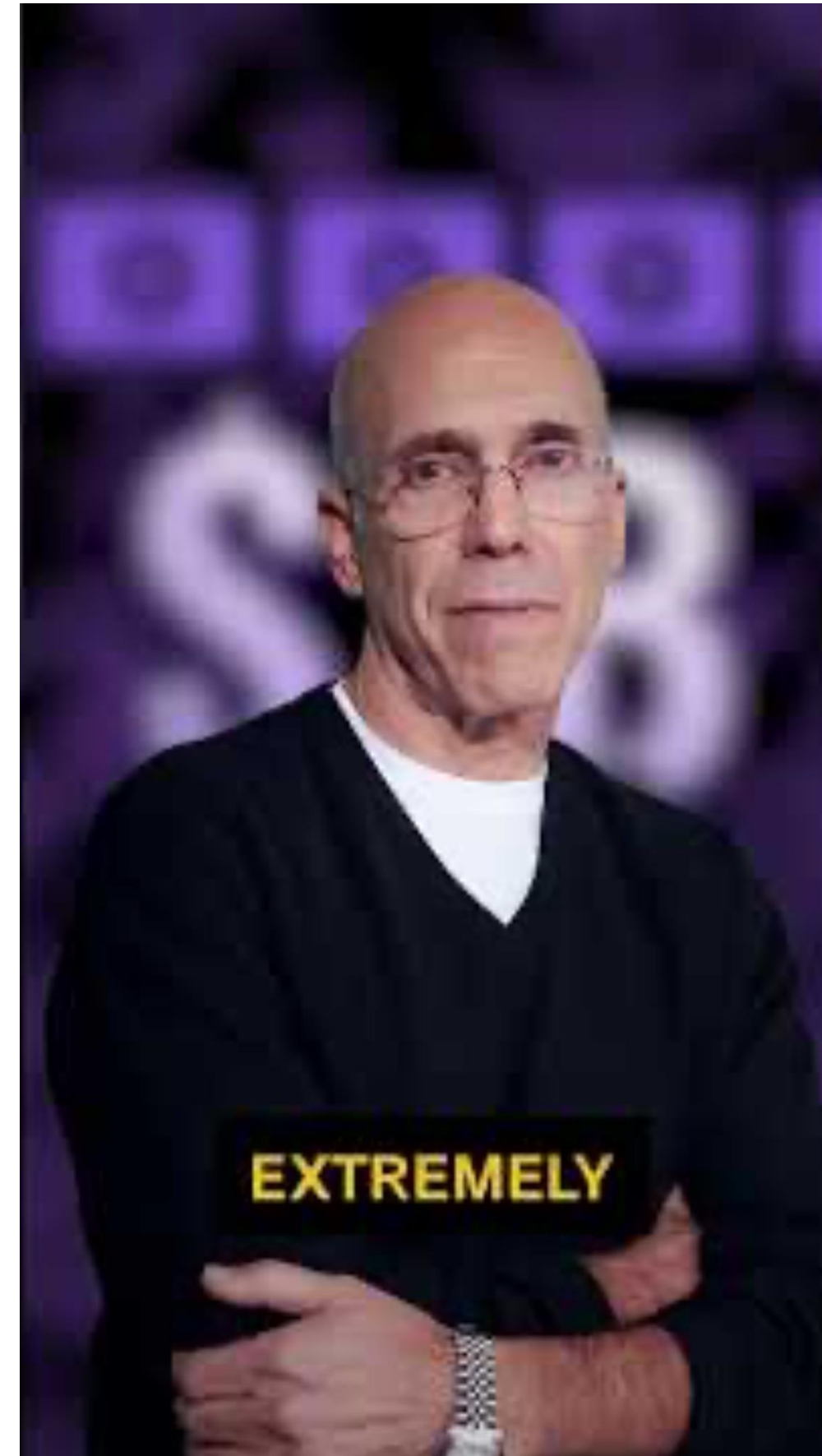


Why startups fail?

And Eventually Build this ...



Let us look at Quibi



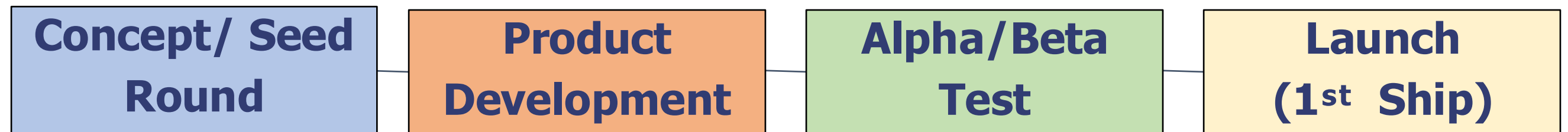
Ref:
<https://www.youtube.com/shorts/rJyqYxeQtY0>

Remember this from EIE1?

Top reasons startups fail



Conventional Approach: Product Development Model



Conventional Approach: Product Development Model

Marketing	Create Market Collateral Create Positioning	Hire PR Agency Early Buzz	Create Demand Launch Event “Branding”
Sales		Hire Sales VP Hire 1 st Sales Staff	Build Sales Organization
Business Development		Hire First Bus Dev	Do deals for for/past launch

Does this Model work? When? Why? Why Not?

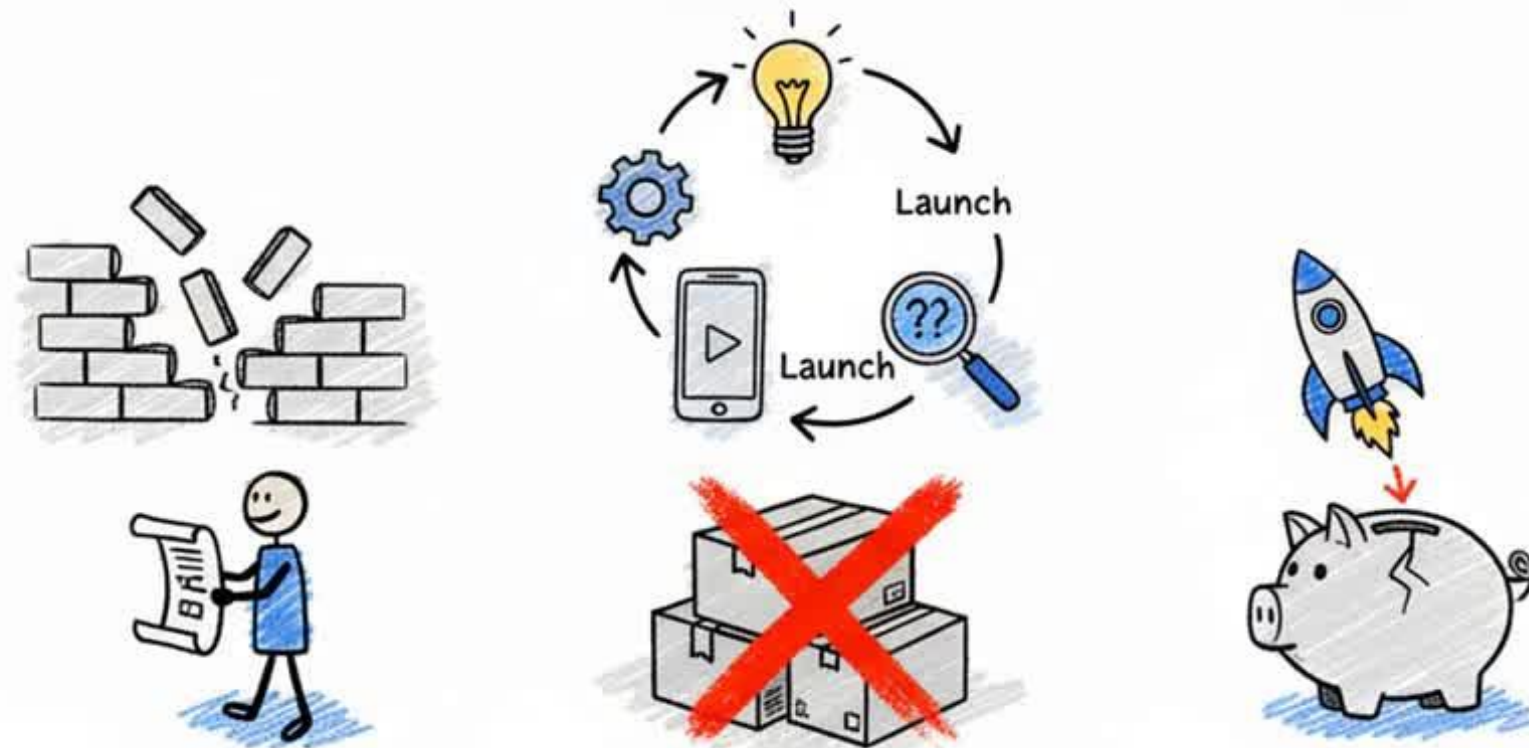
Conventional Approach: Product Development Model

- Business Plan First (BIG assumption: Customers + Market available/ready)
- Startups $\neq$ Smaller version of large companies
- Linear Process
(Idea -> Team -> Product -> Market/Sell -> etc)
- Product Development First (Tech above all else)
- Followed by Test (alpha/beta)



Conventional Approach: Product Development Model

Conventional Product Model



More Startups Fail from a **Lack of
Customers** than from
a Failure of Product Development

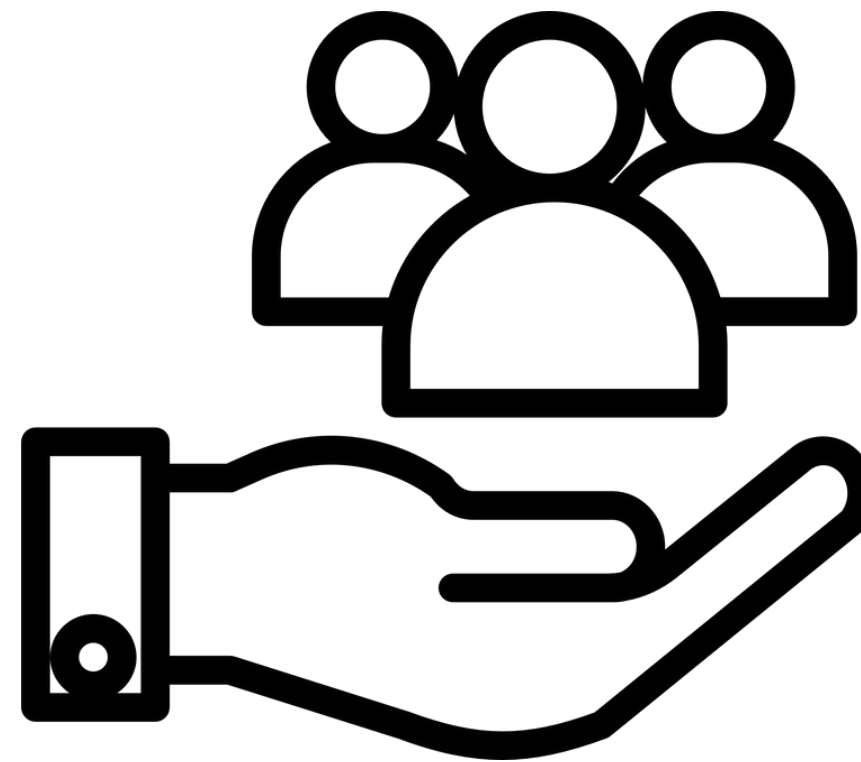


How Do You Fix This?

Focus on Customers and Markets from 'Day 0'

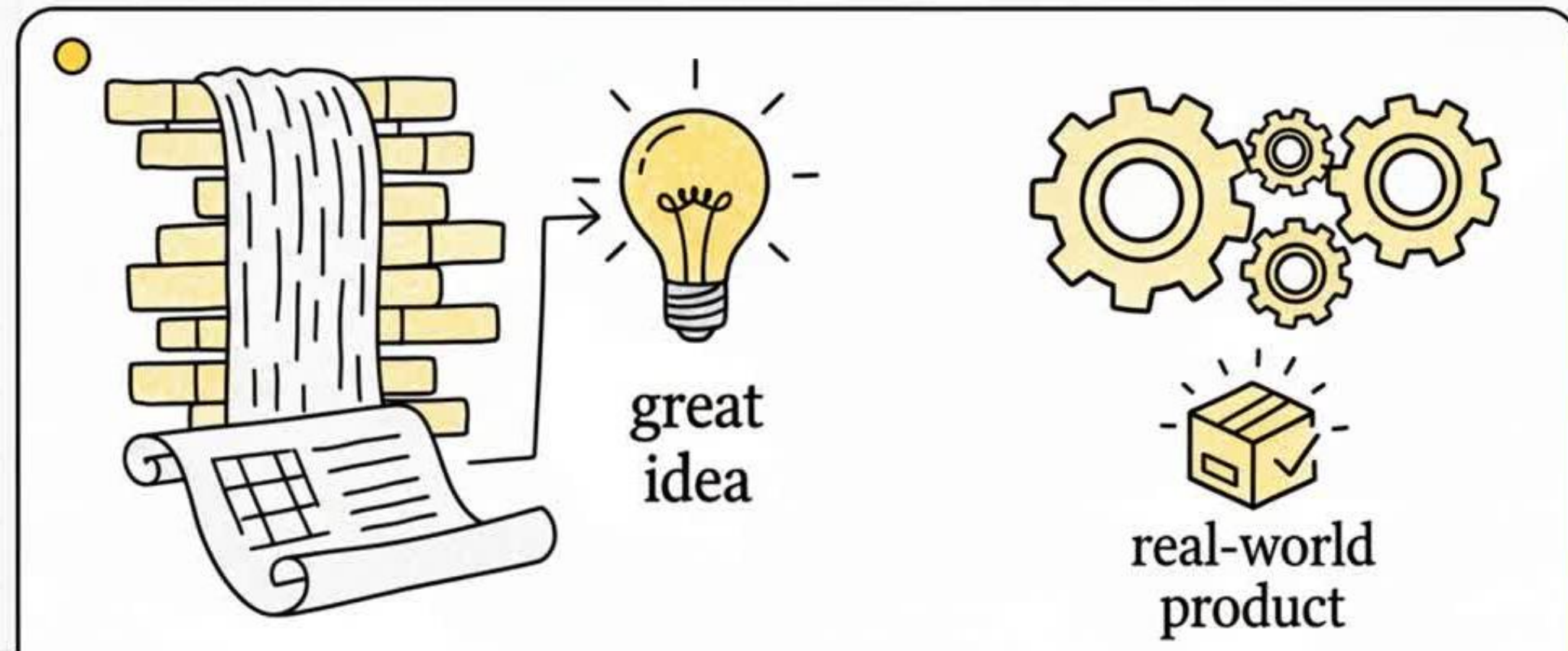
by adopting

'Lean Startup Method'



Required Reading/Viewing

Classic Startup Playbook



Required Reading/Viewing

- **The Lean Approach – Introduction**

(<https://youtu.be/vyR1JjS1Duc>) (Steve Blank, 1:05 mins)

- **The Lean Method**

(<https://youtu.be/GPQExuB-lWw>) (Steve Blank, 5:20 mins)

- **Customer vs Product Development - How to Build a Startup, Steve Blank**

(https://www.youtube.com/watch?v=f_LNNnNfpp4&feature=youtu.be)

Lean Startup Components

**Business
Model
Canvas
(BMC)**

+

**Customer
Development**

+

Agile

Lean Startups Principles

THIS

- 1st Step: Customer Development
- Startup = Temporary Search Organization
- Iterative Process
- Minimum Viable Product (MVP)
- Is there a viable Business Model? Scalable?

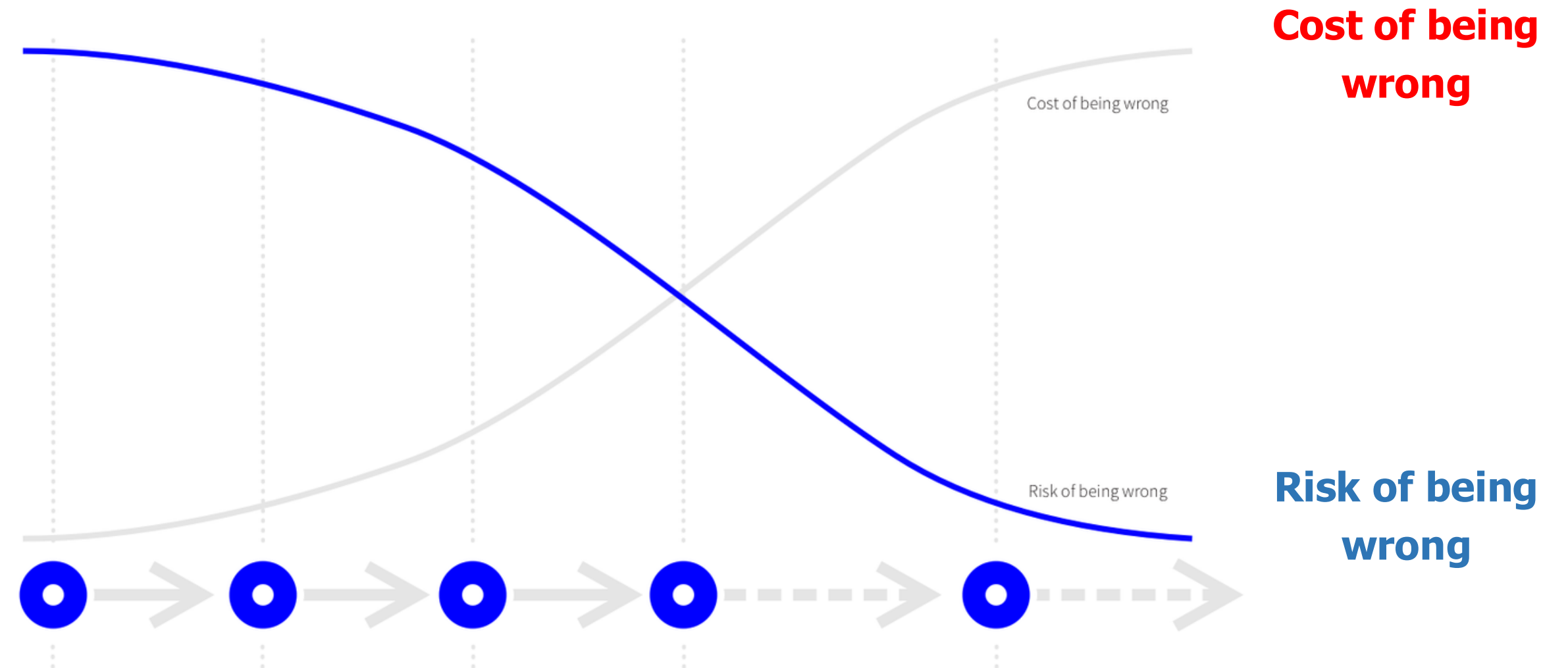
~~NOT THIS~~

- ~~Business Plan First~~
- ~~Startups = Small Companies~~
- ~~Linear Process~~
- ~~Product Development First~~
- ~~Followed by Beta Test~~



Why is this called 'Lean'?

'Lean' (opposite of wasteful) attempts to minimize avoidable 'waste' by identifying risks upfront



Cost vs Risk: '*Cost of being wrong*' keeps growing over time

Lean Startup Components - BMC

Business Model Canvas (BMC) – Prototyping (& NOT Planning) Tool

Key Aspects of BMC

- Desirability:

- Customer Value Proposition, Customer Segments, Customer Relationship, Channel

- Feasibility:

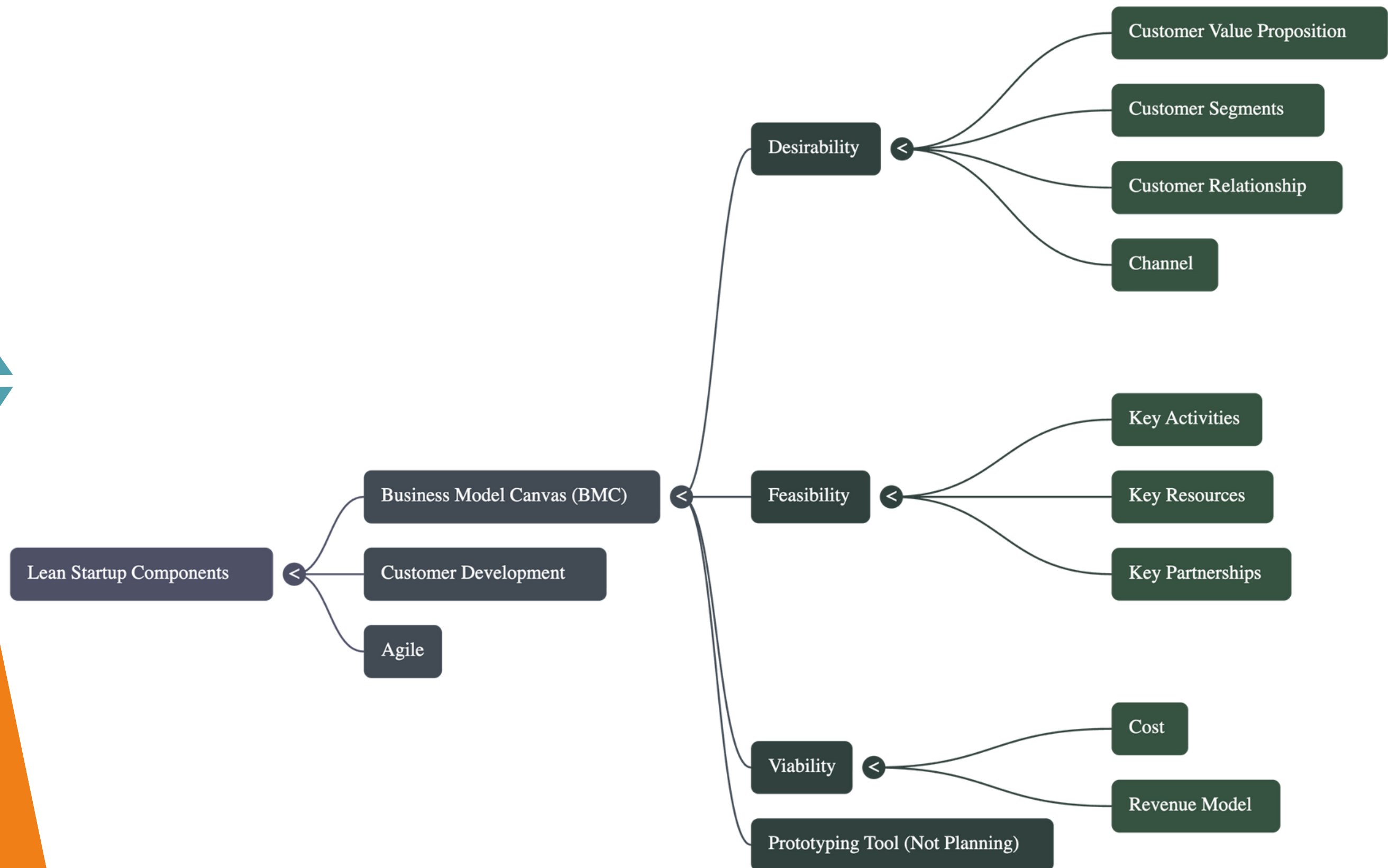
- Key Activities, Key Resources, Key Partnerships

- Viability:

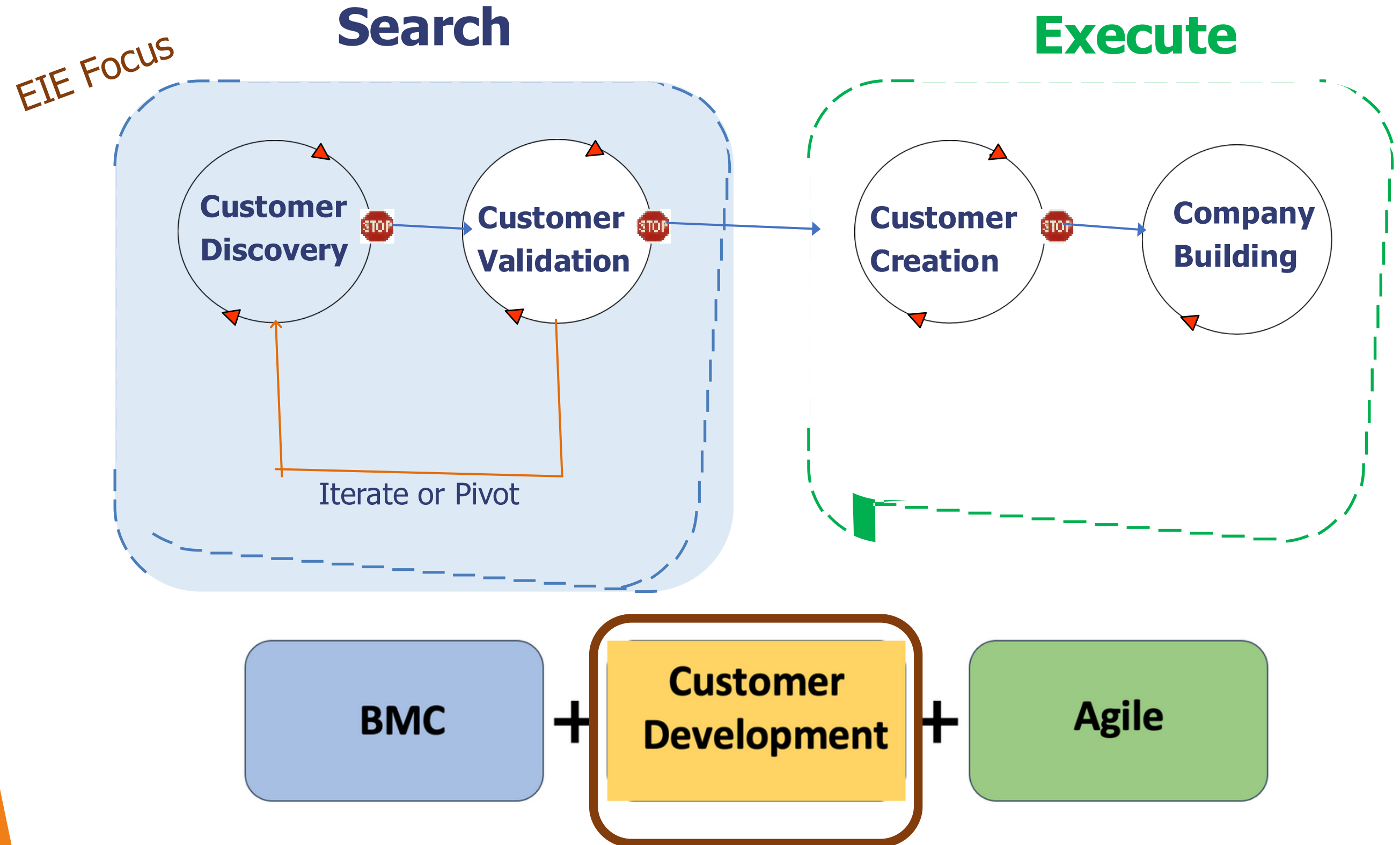
- Cost, Revenue Model



Lean Startup Components - BMC



Lean Startup Components - Customer Development Process



Customer Development Process – Key Points

- Emphasis is on learning & discovery before execution
- Customer Dev: **Parallel process** to Product Development
- Measurable Checkpoints
- Not tied to product schedule, but to Customer Milestones
- Market Type dictates Customer Development process
- “There are no facts inside your building, so get outside”



Agile Development

"Our highest priority is to satisfy the customer through early and continuous delivery of valuable software."

<http://agilemanifesto.org>

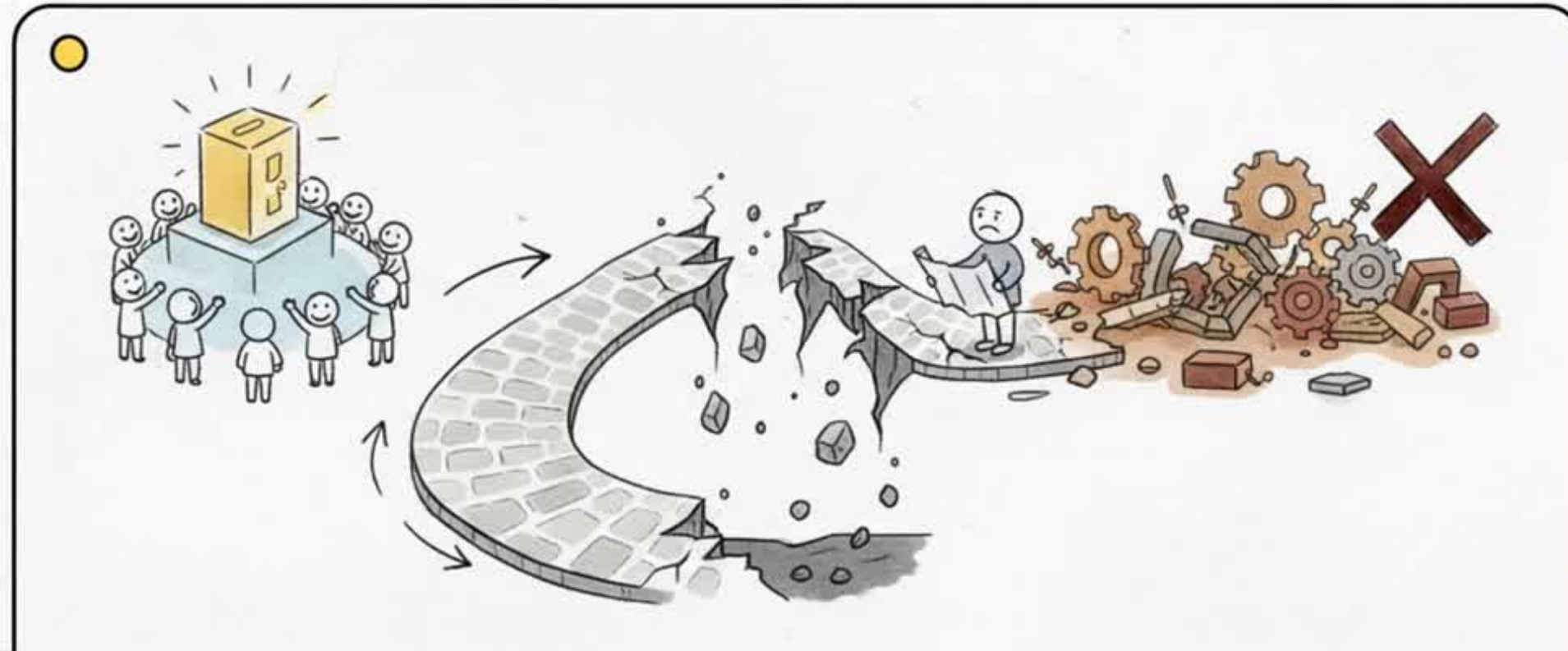


Lean Startup Components – Agile Development

- Prefer flexibility to perfection
 - Ship early and often
 - Test-driven to find and prevent bugs
 - Continuous improvement vs. ship-and-maintain
- Embrace Change
 - Build what you need 'today' (customer-centric)
 - Process-oriented development, so change is relatively painless

Lean Startup Components – Agile Development

The Conventional Product Fallacy



Successful Lean Startups

- A popular team communication and collaboration platform, started as a side project within a game development company called Tiny Speck.
- The founders, Stewart Butterfield, Eric Costello, Cal Henderson, and Serguei Mourachov, initially built an internal tool for their game development team.
- However, they soon realized the potential of the tool as a standalone product and pivoted to focus on building Slack.
- They continuously iterated based on user feedback and market demand, leading to its eventual success.



Successful Lean Startups

- The visual discovery and bookmarking platform, employed lean startup principles in its early stages.
- The founder, Ben Silbermann, started by creating a basic prototype and shared it with a small group of friends to gather feedback.
- He iterated on the product based on user insights, focusing on building a simple and intuitive experience.
- The lean approach helped Pinterest gain traction and refine its features to meet user needs.



Summary

- **Customer-Centric Approach:** Prioritise customer needs over product development. Begin with Customer Development to understand and validate your market.
- **Iterative Progress:** Embrace an iterative process:
 - Idea → Team → Product → Market; continuously improve and adapt based on feedback.
- **Minimum Viable Product (MVP):** Build the simplest version of your product that validates your business model.
 - Test assumptions efficiently.
- **Fail Fast, Learn Faster:** Don't fear failure; it's a stepping stone.
 - Focus on learning from failures and pivoting if necessary.
- **Continuous Improvement:** Embrace change and agility.
 - Develop a process-oriented mindset that fosters painless adaptation.
- **Take Action Today:** Embrace Lean Startup principles to validate your idea, satisfy customer needs, and build a scalable business model. Iterate, adapt, and transform your startup journey.

Module 1: Post Session Brief

**Centre for Innovation and
Entrepreneurship (CIE)**

CIE LEVEL 2 - Building a Lean Startup Required Task - 1

Activity

- Reflect on what kind of teammates you are looking for?
- Scout for potential teammates (each team will have 5-6 members)
- Be ready to discuss what role fits you best (Mkt, Tech, Biz, Prototyping, etc)
 - Opportunity theme(s):
 - What do you/team like/excited about?
 - Is “Desirable – Feasible – Viable”?
 - Can be completed (course requirement) in ~10 wks?